

3 The photographs show plants growing in vertical and horizontal systems.

In both of these systems, the plants grow in nutrient solutions instead of in soil.



vertical system

(Source: © Lano Lan/Alamy Stock Photo)



horizontal system

(Source: © RossHelen editorial/Alamy Stock Photo)

(a) (i) The nutrient solutions contain magnesium ions and nitrates.

Describe the roles of these inorganic ions in the plants.

(2)

Magnesium ions

.....

Nitrates

.....

(ii) Suggest **one** advantage of the vertical system over the horizontal system.

(1)

.....

.....

.....

(b) Fruits contain cells.

The effect of calcium ion concentration on the firmness of fruits was investigated.

Cherry plants were treated once, twice or three times, with two different concentrations of calcium chloride solution. The firmness of the cherry fruits from these plants was then measured.

The table shows the effect on the mean fruit firmness of these treatments and a control.

Number of treatments	Mean fruit firmness / N cm ⁻²	
	Plants treated with a calcium chloride solution of 950 mg dm ⁻³	Plants treated with a calcium chloride solution of 3 800 mg dm ⁻³
0 (control)	4.03	4.03
1	4.08	4.18
2	4.09	4.28
3	4.19	4.61

- (i) Which is the percentage increase in mean fruit firmness between the cherries from the control plants and those from plants treated three times with the higher calcium chloride concentration?

(1)

- ☐ **A** 3.72 %
- ☐ **B** 3.97 %
- ☐ **C** 6.20 %
- ☐ **D** 14.39 %



(ii) Give **one** conclusion for this investigation.

(1)

(iii) Explain the effect of calcium ion concentration on the firmness of cherry fruits.

(3)

(Total for Question 3 = 8 marks)

